Plan - SketchImpact

Group 2

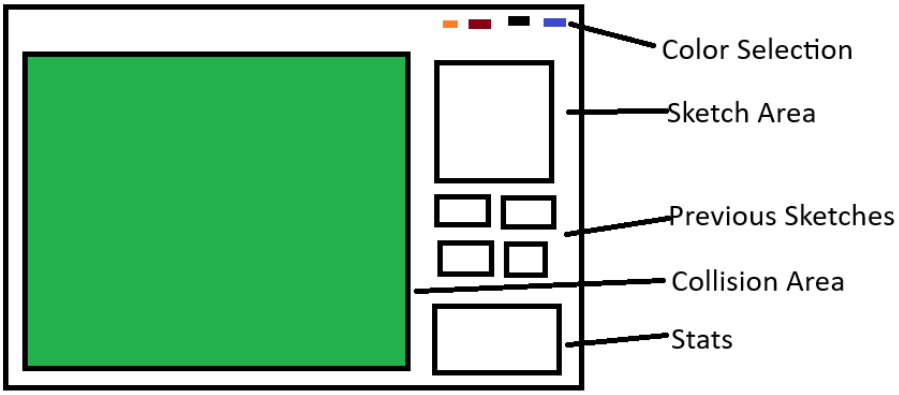
- Rene Sebastian
- Shabaz Malik
- Frederick David Vandermoeten

Problem Statement

We have highly optimized techniques for collision detection between geometric primitives such as circles, rectangles or polygons. However, for user-generated freeform sprites, the narrow phase becomes expensive. We try to implement a lightweight AI model that predicts collisions between sprite pairs to filter the amount of collisions we have to check.

Game

The player can draw their own sketches and watch them collide on a 2D billiard table. The player can influence their sketches speed by dragging/ clicking with the mouse. There is visual feedback showing stats such as the number of skipped collision checks.



Techniques

- Broad and Narrow Collision detection
- some Spatial Data Structure
- AABB/ OBB
- NN for filtering before narrow phase, or to replace narrow phase

Workflow

1. Build baseline game without AI

2. Generate labeled dataset of sprite pairs, relative offsets and collision outcomes
3. Train a CNN model
4. Integrate it into the game
5. Evaluate performance and accuracy